

Thyroid-stimulating Hormone (TSH)

Test Details

METHODOLOGY

Electrochemiluminescence immunoassay (ECLIA)

RESULT TURNAROUND TIME

Within 1 day

Turnaround time is defined as the usual number of days from the date of pickup of a specimen for testing to when the result is released to the ordering provider. In some cases, additional time should be allowed for additional confirmatory or additional reflex tests. Testing schedules may vary.

Use

Thyroid function test. Investigation of low thyroxine (T_4) result; the differential diagnosis of primary hypothyroidism from normal, and the differential diagnosis of primary hypothyroidism from pituitary/hypothalamic hypothyroidism. TSH is high in primary hypothyroidism. Low TSH occurs in hyperthyroidism. Evaluation of therapy in hypothyroid patients receiving various thyroid hormone preparations: Low values are found in states of excessive thyroid replacement. Normal result on a sensitive TSH assay is acceptable evidence of adequate thyroid replacement.

Follow-up of patients who have had hyperthyroidism treated with radioiodine or surgery. Follow-up low T_4 newborn results.

This third-generation TSH assay can be considered a test for thyroid disease. A result within the accepted reference interval provides strong evidence for euthyroidism.

Special Instructions

This test may exhibit interference when sample is collected from a person who is consuming a supplement with a high dose of biotin (also termed as vitamin B7 or B8, vitamin H, or coenzyme R). It is recommended to ask all patients who may be indicated for this test about biotin supplementation. Patients should be cautioned to stop biotin consumption at least 72 hours prior to the collection of a sample.

Limitations

Spurious increase from antiovine TSH antibodies by double-antibody technique has been reported.² TSH may be affected by glucocorticoids, dopamine, and by severe illness,³ and these remain limitations even for the new, sensitive TSH assays. TSH suppression in hypothyroidism with severe illness has been reported with TSH increase with recovery.⁴ Normal TSH levels in the presence of hypothyroidism have been reported with head injury.⁵ Iopanoic acid, ipodate, and an antiarrhythmic drug, amiodarone, cause changes in thyroid test results including increases in T_4 , free T_4 , and TSH and decreases of T_3 .⁶ TSH is not elevated in secondary hypothyroidism.

Probably no single test, even the sensitive immunoassays, can be expected to adequately reflect thyroid status under all circumstances. Among possible problems are the recovery phase of nonthyroidal illness, states of resistance to thyroid hormone, thyrotropin-producing tumors, thyroid status in acute psychiatric illness, early in thyrotoxicosis and in subacute thyroiditis.⁷

Custom Additional Information

The consequences of subclinical thyroid disease (serum TSH 0.1–0.45 $\mu\text{IU/mL}$ or 4.5–10.0 $\mu\text{IU/mL}$) are minimal and current guidelines recommend against routine treatment of patients with TSH levels in these ranges, but thyroid function tests should be repeated at 6- to 12-month intervals to monitor TSH levels;⁸ however, treatment of subclinical hypothyroidism is indicated in patients with TSH levels $>10.0 \mu\text{IU/mL}$ or in patients with TSH levels $<10.0 \mu\text{IU/mL}$ in conjunction with goiter or positive for antithyroid peroxidase antibodies (or both).⁹ In patients who are receiving replacement therapy, the dose should be adjusted so serum TSH values range from 0.3–3.0 $\mu\text{IU/mL}$. An exception is thyroid hormone replacement treatment after thyroidectomy for differentiated thyroid cancer, in which case, a mildly to moderately suppressed TSH level is generally desirable.¹⁰ It is reasonable to consider serum TSH measurement for pregnant women or women planning to become pregnant with a family history of thyroid disease, prior thyroid dysfunction, symptoms or physical findings suggestive of hypo- or hyperthyroidism, an abnormal thyroid gland on examination, type 1 diabetes mellitus, or a personal history of autoimmune disorder.¹¹ Suggested upper limit for the TSH reference range for pregnant women and preconception is: first trimester – $<2.5 \mu\text{IU/mL}$, and $3.0 \mu\text{IU/mL}$ in the second and third trimesters.¹⁰

Unsuspected increase in the level of serum TSH is not uncommon in elderly subjects. A study by Sawin et al found that 22 of 344 (5.9%) healthy persons older than age 60 had a TSH level $>10 \mu\text{IU/mL}$; 10 of the 22 had low T_4 and FT_4 index. Elderly hypothyroid individuals may have minimal recognizable clinical symptoms of thyroid deficiency.¹¹ TSH is the single most sensitive test for primary hypothyroidism. If there is clear evidence for hypothyroidism and the TSH is not elevated, hypopituitarism should be considered (secondary hypothyroidism).

TSH levels have been elevated or inappropriately detectable for high thyroid hormone levels in some patients with thyrotropin-secreting pituitary adenomas. Delay in diagnosis of these tumors may lead to visual compromise. The effects of such neoplasms can be misdiagnosed as those of primary hyperthyroidism.

Until the late 1980s, TSH assays were not sufficiently sensitive to distinguish hyperthyroidism from euthyroid (normal) subjects. The new generation of ultrasensitive TSH immunoassays have provided a far more effective diagnostic separation of thyrotoxicosis from euthyroidism.

This assay has a sensitivity of 0.004 μ IU/mL and meets all criteria as a third-generation TSH assay.

Specimen Requirements

Specimen

Serum (preferred) **or** plasma

Volume

1 mL

Minimum Volume

0.7 mL (**Note:** This volume does **not** allow for repeat testing.)

Container

Red-top tube, gel-barrier tube, **or** green-top (lithium heparin) tube. Do **not** use oxalate, EDTA, or citrate plasma.

Collection Instructions

If a red-top tube or plasma is used, transfer separated serum or plasma to a plastic transport tube.

Stability Requirements

Temperature	Period
Room temperature	14 days
Refrigerated	14 days
Frozen	14 days
Freeze/thaw cycles	Stable x3

Reference Range

See table.¹

Age	Range (μIU/mL)
0 to 6 d	0.700–15.200
7 d to 3 m	0.720–11.000
3 m 1 d to 12 m	0.730–8.350
1 to 5 y	0.700–5.970
6 to 10 y	0.600–4.840
>10 y	0.450–4.500

Storage Instructions

Room temperature

Causes for Rejection

Citrate plasma specimen; improper labeling

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Footnotes

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